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practical 4 :

4.1.1

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

**Input Format:**

- The user should enter a list of numbers separated by space when prompted.

**Output Format:**

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.  
code and output :

```
1 import pandas as pd
2
3 # Take inputs from the user to create a list of numbers
4 numbers = list(map(int, input().split()))
5
6 # Create a Pandas series from the list of numbers
7 series = pd.Series(numbers)
8 # Grouping by even and odd numbers and calculating the mean
9 grouped = series.groupby(series % 2 == 0).mean()
10
11 # Display the mean of even and odd numbers with labels
12 grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
13 print("Mean of even and odd numbers:")
14 print(grouped)
15
```

Average time: 0.010 s, 9.67 ms | Maximum time: 0.025 s, 25.00 ms | 3 out of 3 shown test case(s) passed | 3 out of 3 hidden test case(s) passed

Test case 1 (25 ms) [Debug] [Test cases]

| Expected output               | Actual output                 |
|-------------------------------|-------------------------------|
| 1 2 3 4 5 6 7 8 9 10          | 1 2 3 4 5 6 7 8 9 10          |
| Mean of even and odd numbers: | Mean of even and odd numbers: |
| Odd: ... 5.0                  | Odd: ... 5.0                  |
| Even: ... 6.0                 | Even: ... 6.0                 |
| dtype: float64                | dtype: float64                |

Terminal | Test cases

#### 4.1.2

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

#### Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

#### Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

#### Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

#### Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

### Add a new column:

|  |  |
|--|--|
|  |  |
|  |  |

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

**Modify a column:**

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

|  |  |
|--|--|
|  |  |
|  |  |
|  |  |

### Delete a column:

- Remove the **Age** column.
- Display the DataFrame after deleting the column.

### Sample Test Cases

- dataframeOps.py

v

```
# import pandas as pd
```

```
# df = pd.DataFrame(data)
```

```
# # Display the original DataFrame
```

```
# print("Original DataFrame:")
```

```
# print(df)
```

```
# # Adding a new row
```

```
# # Display the DataFrame after adding a new row
```

```
# print("After adding a row:\n",df)
```

```
# # Modifying a row
```

```
# # Display the DataFrame after modifying a row
```

```
# print("After modifying a row:")
```

```
# print(df)

# # Deleting a row

# # Display the DataFrame after deleting a row
# print("After deleting a row:")
# print(df)

# # Adding a new column

# # Display the DataFrame after adding a new column
# print("After adding a new column:")
# print(df)

# # Modifying a column

# # Display the DataFrame after modifying a column
# print("After modifying a column:")
# print(df)

# # Deleting a column

# # Display the DataFrame after deleting a column
# print("After deleting a column:")
# print(df) import pandas
as pd

# Provided dictionary of lists data
```

```
= {  
'Name': ['Alice', 'Bob', 'Charlie'],  
'Age': [25, 30, 35],  
}
```

```
# Convert the dictionary to a DataFrame df  
= pd.DataFrame(data)
```

```
# Display the original DataFrame print("Original  
DataFrame:") print(df)
```

```
# Adding a new row new_name=input("New name:  
") new_age=int(input("New age: "))  
new_row={'Name':new_name,'Age':new_ag  
e}  
df=pd.concat([df,pd.DataFrame([new_row])  
,ignore_index=True)  
# Display the DataFrame after adding a new row print("After adding  
a row:\n",df)
```

```
# Modifying a row modify_index=int (input("Index of  
row to modify: ")) new_age_mod=int(input  
("New age: ")) df.loc[modify_index,"Age"]=new_age_mod
```

```
# Display the DataFrame after modifying a row print("After  
modifying a row:") print(df)
```

```
# Deleting a row delete_index=int(input("Index of  
row to delete: "))
```

**Translation Failed**

**Reason for late submission**

Please enter at least 15 characters

code :

```
import pandas as pd
```

```
# Provided dictionary of lists data
```

```
= {  
    'Name': ['Alice', 'Bob', 'Charlie'],  
    'Age': [25, 30, 35],  
}
```

```
# Convert the dictionary to a DataFrame df
```

```
= pd.DataFrame(data)
```

```
# Display the original DataFrame print("Original
```

```
DataFrame:") print(df)
```

```
# Adding a new row new_name=input("New name: ")
```

```
new_age=int(input("New age: "))
```

```
new_row={'Name':new_name,'Age':new_age}
```

```
df=pd.concat([df,pd.DataFrame([new_row]),ignore_index=True) #
```

```
Display the DataFrame after adding a new row print("After adding
```

```
a row:\n",df)
```



```
# Modifying a row modify_index=int (input("Index of  
row to modify: ")) new_age_mod=int(input  
("New age: ")) df.loc[modify_index,"Age"]=new_age_mod
```

```
# Display the DataFrame after modifying a row print("After  
modifying a row:") print(df)
```

```
# Deleting a row delete_index=int(input("Index of  
row to delete: "))  
df=df.drop(delete_index).reset_index(drop=True)
```

```
# Display the DataFrame after deleting a row print("After  
deleting a row:") print(df)
```

```
# Adding a new column gender_input=input("Enter genders  
separated by space: ") genders=gender_input.split()  
df["Gender"]= genders
```

```
# Display the DataFrame after adding a new column print("After  
adding a new column:") print(df)
```

```
# Modifying a column df["Name"]=df["Name"].str.upper()
```

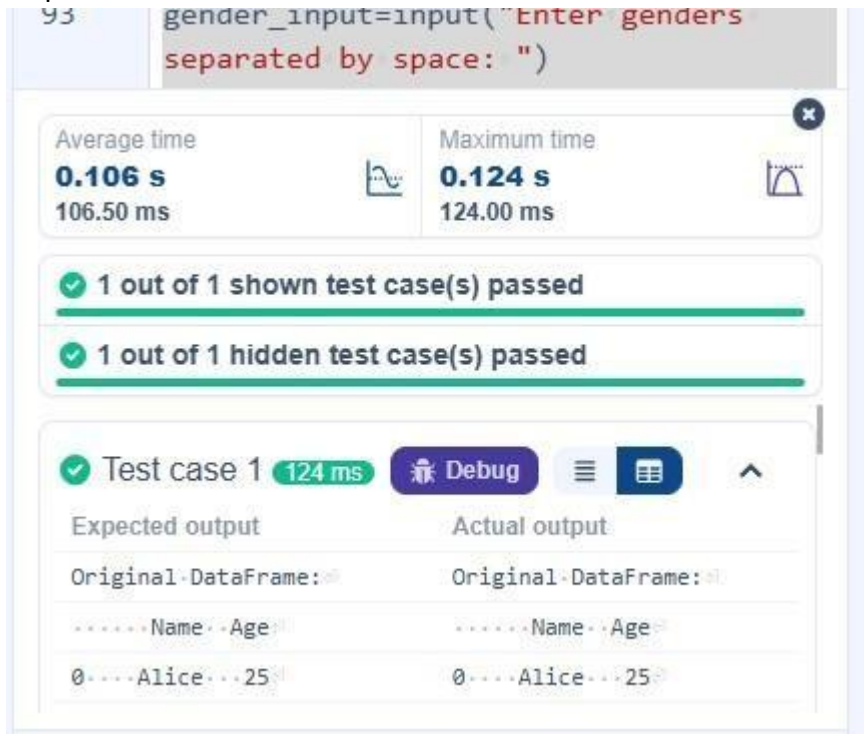
```
# Display the DataFrame after modifying a column  
print("After modifying a column:") print(df)
```

```
# Deleting a column df=df.drop(columns=['Age'])
```

```
# Display the DataFrame after deleting a column print("After deleting  
a column:")
```

```
print(df)
```

output :



#### 4.1.3

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

#### Note:

Refer to the displayed test cases for better understanding.

Code :

```
import pandas as pd
```

```
# Read the text file into a DataFrame file = input() data = pd.read_csv(file,
sep="\s+", header=None, names=["Name", "Age", "Grade"])

# Display the first five rows print("First
five rows:") print(data.head())

# Calculate the average age (rounded to 2 decimal places)
average_age = round(data['Age'].mean(), 2) print(f"Average age:
{average_age}")

# Filter students with grade up to threshold 'B' grade_order = {'A': 1, 'B': 2, 'C':
3, 'D': 4, 'F': 5} threshold = 'B' filtered_df = data[data['Grade'].map(grade_order)
<= grade_order[threshold]]

print(f"Students with a grade up to {threshold}")
print(filtered_df) output
:
```

```
20     to 2 decimal places)
    average_age =
21     round(data['Age'].mean(), 2)
    print(f"Average age: {average_age}")
```

Average time  
**0.047 s**  
47.00 ms

Maximum time  
**0.068 s**  
68.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 1 out of 1 hidden test case(s) passed

✓ Test case 1 68 ms

#### 4.2.1

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

#### Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago  
2025-01-03,Product B,2,15,Chicago  
2025-01-03,Product A,8,20,Los Angeles  
2025-01-04,Product C,6,30,New York  
2025-01-04,Product B,5,15,Los Angeles  
2025-01-05,Product A,3,20,Chicago  
2025-01-05,Product C,10,30,Los Angeles

**Note:**

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code :

```
import pandas as pd
```

```
# Prompt the user for the file name file_name
```

```
= input()
```

```
# Load the data df = pd.read_csv(file_name)
```

```
# Convert 'Date' to datetime df['Date']
```

```
= pd.to_datetime(df['Date'])
```

```
# Calculate total sales for each row df['TotalSales']
```

```
= df['Quantity'] * df['Price']
```

```
# Extract month in YYYY-MM format df['Month']
```

```
= df['Date'].dt.to_period('M') # Group by month
```

```
and sum total sales monthly_sales = df.groupby('Month')['TotalSales'].sum()
```

```
# Find the month with highest total sales best_month
= monthly_sales.idxmax() highest_sales
= monthly_sales.max()

print(f"Best month: {best_month}") print(f"Total sales:
${highest_sales:.2f}")
```

output :

Average time  
**0.030 s**  
29.67 ms

Maximum time  
**0.061 s**  
61.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 61 ms Debug

| Expected output        | Actual output          |
|------------------------|------------------------|
| sales_data.csv         | sales_data.csv         |
| Best month: 2025-01    | Best month: 2025-01    |
| Total sales: \$1210.00 | Total sales: \$1210.00 |

< Prev Reset Submit Next >

#### 4.2.2

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

**Sample Data:**

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

**Note:**

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code :

```
import pandas as pd
```

```
# Prompt the user for the file name file_name
```

```
= input()
```

```
# Load the CSV data df = pd.read_csv(file_name)
```

```
# Group by Product and sum the Quantity
```

```
product_sales
```

```
= df.groupby('Product')['Quantity'].sum()
```

```
# Find the product with the highest quantity sold best_product
```

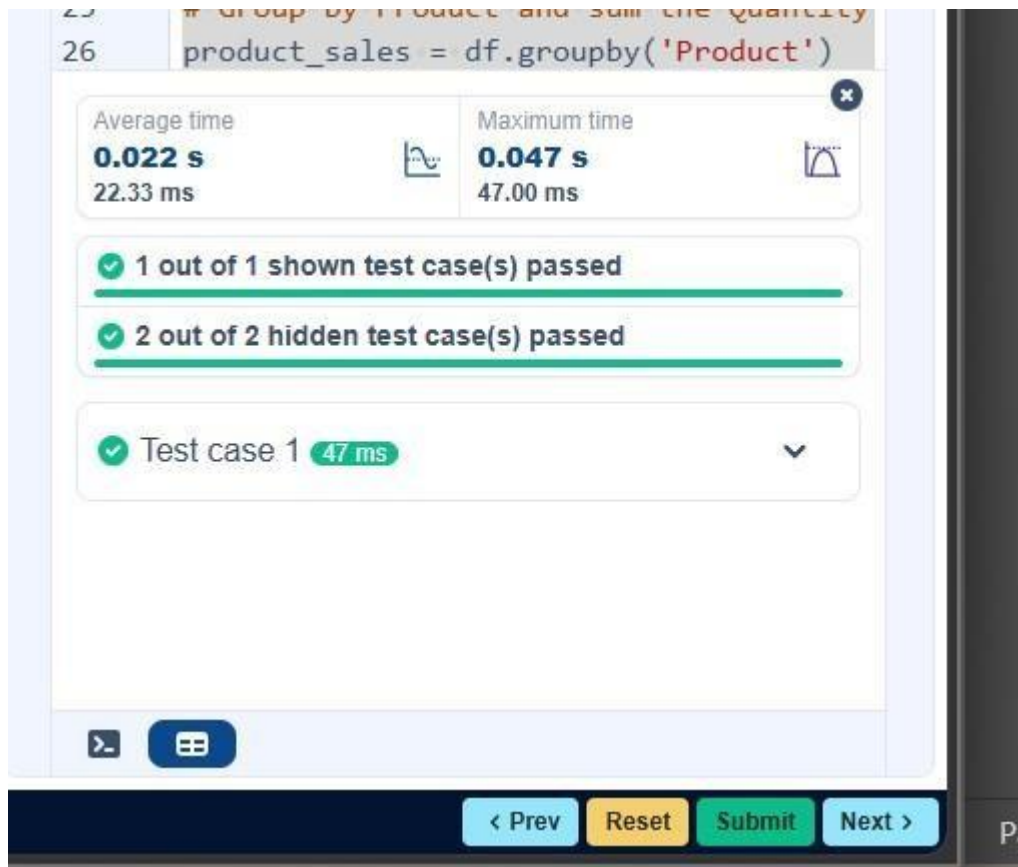
```
= product_sales.idxmax() highest_quantity
```

```
= product_sales.max()
```

```
# Display the result print(f"Best selling
```

```
product: {best_product}") print(f"Total
```

```
quantity sold: {highest_quantity}") output :
```



#### 4.2.3

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

**Sample Data:**



Date,Product,Quantity,Price,City  
2025-01-01,Product A,5,20,New York  
2025-01-01,Product B,3,15,Los Angeles  
2025-01-02,Product A,7,20,New York  
2025-01-02,Product C,4,30,Chicago  
2025-01-03,Product B,2,15,Chicago  
2025-01-03,Product A,8,20,Los Angeles  
2025-01-04,Product C,6,30,New York  
2025-01-04,Product B,5,15,Los Angeles  
2025-01-05,Product A,3,20,Chicago  
2025-01-05,Product C,10,30,Los Angeles

**Note:**

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code: import pandas  
as pd

```
# Prompt the user for the file name file_name  
= input()
```

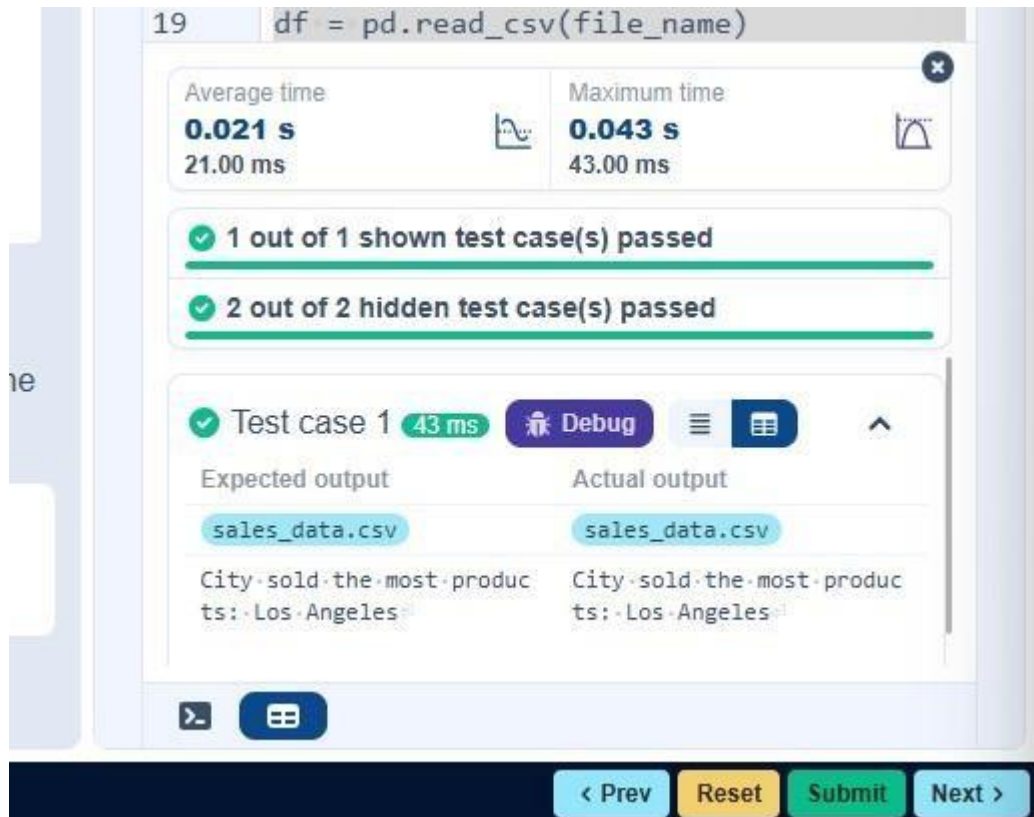
```
# Load the CSV data df = pd.read_csv(file_name)
```

```
# Group by City and sum the Quantity city_sales  
= df.groupby('City')['Quantity'].sum()
```

```
# Find the city with the highest total quantity sold best_city  
= city_sales.idxmax()
```

# Display the result print(f"City sold the most products:

{best\_city}") output :



#### 4.2.4

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

#### Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago 2025-01-05,Product  
C,10,30,Los Angeles

### Explanation:

### Transactions:

- **2025-01-01:** Product A, Product B
- **2025-01-02:** Product A, Product C
- **2025-01-03:** Product B, Product A • **2025-01-04:** Product C, Product B
- **2025-01-05:** Product A, Product C

Now, let's count how often the pairs of products appear together:

- **Product A and Product B:** Appear in transactions on 2025-01-01 and 2025-01-03. •  
**Product A and Product C:** Appear in transactions on 2025-01-02 and 2025-01-05.
- **Product B and Product C:** Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

- **Product A and Product B** (2 times)
- **Product A and Product C** (2 times)

### Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code: import pandas as pd from itertools

import combinations from collections

import Counter

# Prompt user to input the file name file\_name

```
= input()
```

```
# Read data from CSV df = pd.read_csv(file_name)
```

```
# Initialize a list to store all product pairs all_pairs
```

```
= []
```

```
# Group by each date for date, group in
```

```
df.groupby('Date'): products =
```

```
group['Product'].tolist() # Generate all unique
```

```
pairs for this date pairs =
```

```
combinations(sorted(products), 2)
```

```
all_pairs.extend(pairs)
```

```
# Count frequency of each product pair pair_counts
```

```
= Counter(all_pairs)
```

```
# Find the maximum frequency max_freq
```

```
= max(pair_counts.values())
```

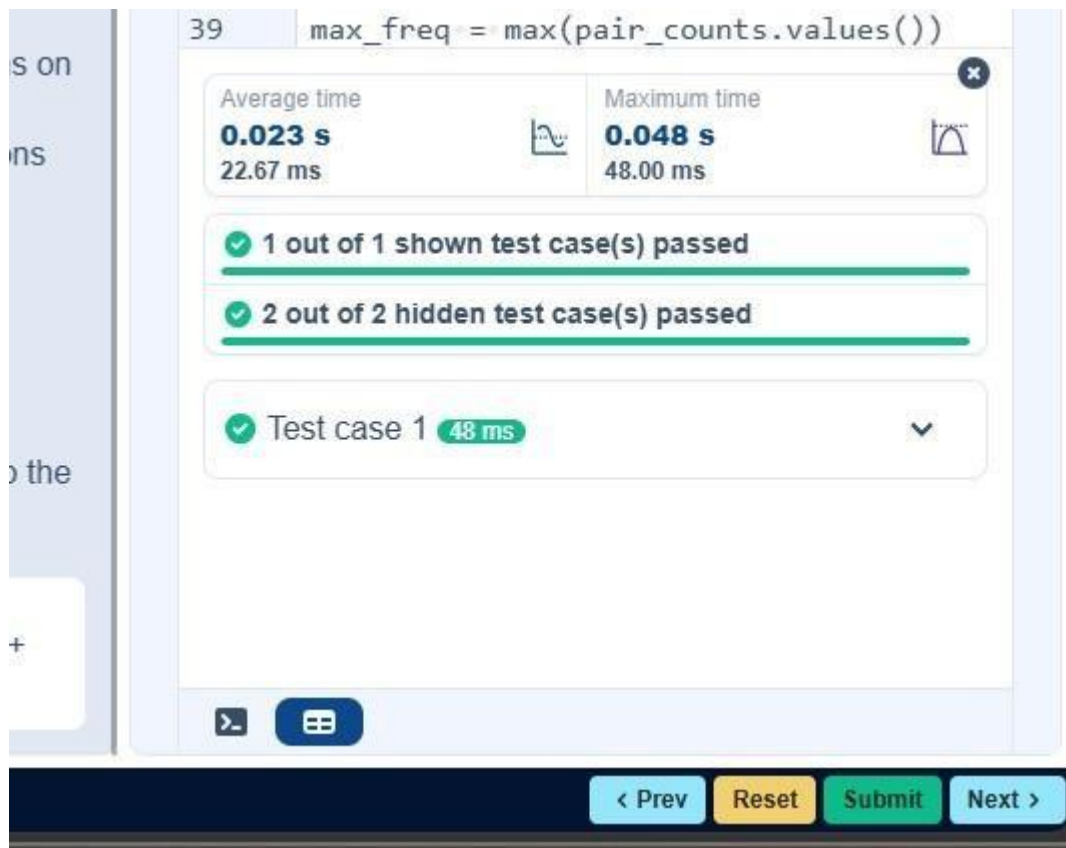
```
# Get all pairs that have the maximum frequency most_frequent_pairs = [pair for pair, count  
in pair_counts.items() if count == max_freq]
```

```
# Display the result in the exact expected format for pair
```

```
in most_frequent_pairs:
```

```
    print(f"{pair[0]} and {pair[1]}: {max_freq} times")
```

```
output:
```



#### 4.2.5

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

### Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC  
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

| PassengerId                                                                  | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|------------------------------------------------------------------------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
| 10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C |          |        |      |     |     |       |       |        |      |       |          |

**Note:** Refer to the visible test case for better reference.

Code :

```
import pandas as pd
import numpy
```

```
as np
```

```
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
```

# 1. Display the first 5 rows of the dataset `print(data.head())`

# 2. Display the last 5 rows of the dataset `print(data.tail())`

# 3. Get the shape of the dataset `print(data.shape)`

# 4. Get a summary of the dataset (info) `print(data.info())`

# 5. Get basic statistics of the dataset `print(data.describe())`

# 6. Check for missing values

`print(data.isnull().sum())`

# 7. Fill missing values in the 'Age' column with the median age `data['Age']`

`= data['Age'].fillna(data['Age'].median())`

# 8. Fill missing values in the 'Embarked' column with the mode `data['Embarked'] =`

`data['Embarked'].fillna(data['Embarked'].mode()[0])`

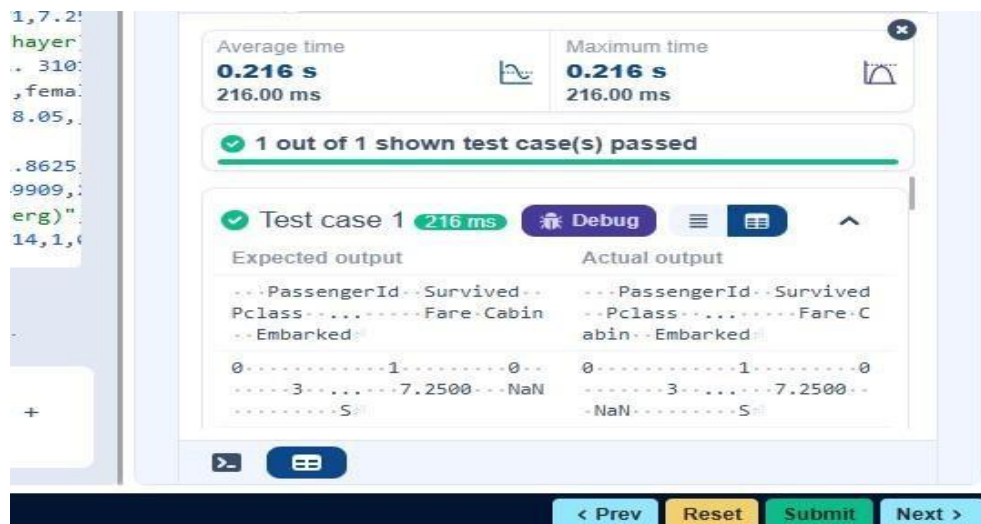
# 9. Drop the 'Cabin' column due to many missing values `data`

`= data.drop(columns=['Cabin'])`

# 10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'

`data['FamilySize'] = data['SibSp'] + data['Parch']` output

:



#### 4.2.6

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|

5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,



### Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

**Note:** Refer to the visible test case for better reference.

```
Code: import pandas
as pd import numpy
as np
```

```
# Load the Titanic dataset data =
pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']

# 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise) data['IsAlone'] =
(data['FamilySize'] == 0).astype(int)

# 2. Convert 'Sex' to numeric (male: 0, female: 1) data['Sex']
= data['Sex'].map({'male': 0, 'female': 1})
```

```
# 3. One-hot encode the 'Embarked' column data = pd.get_dummies(data,  
columns=['Embarked'])
```

```
# 4. Get the mean age of passengers print(data['Age'].mean())
```

```
# 5. Get the median fare of passengers print(data['Fare'].median())
```

```
# 6. Get the number of passengers by class print(data['Pclass'].value_counts())
```

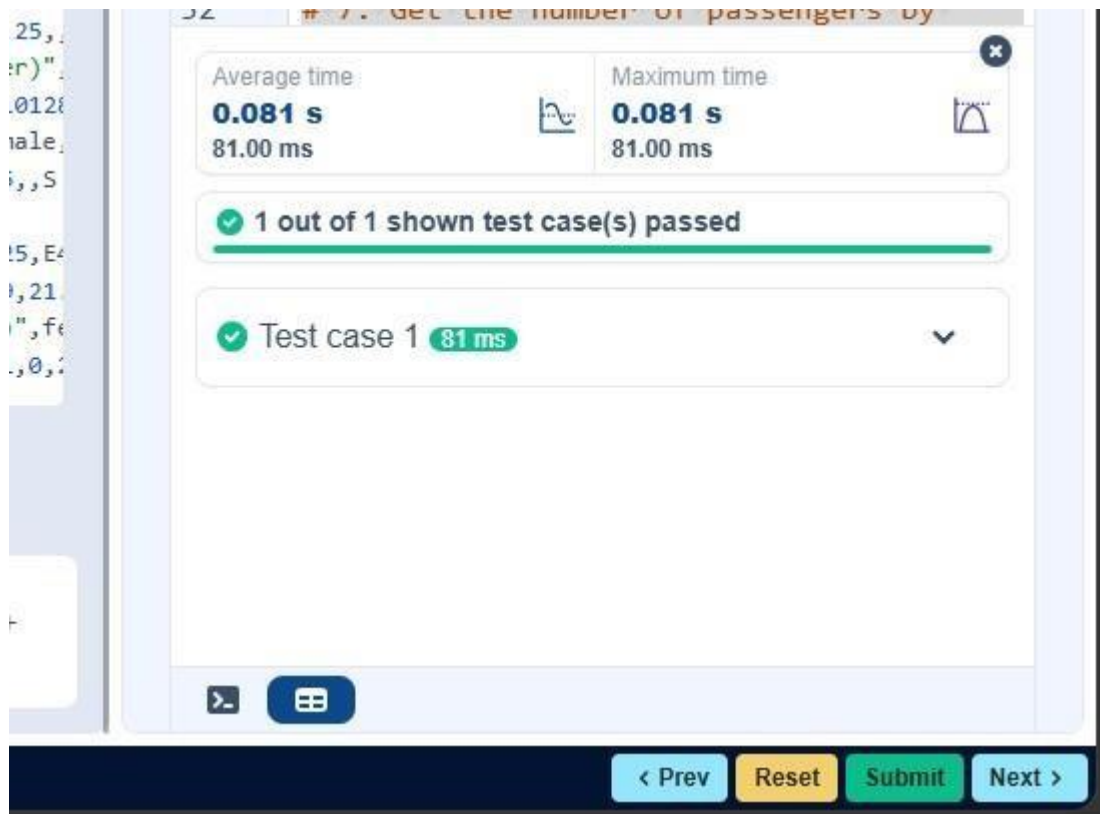
```
# 7. Get the number of passengers by gender print(data['Sex'].value_counts())
```

```
# 8. Get the number of passengers by survival status print(data['Survived'].value_counts())
```

```
# 9. Calculate the survival rate print(data['Survived'].mean())
```

```
# 10. Calculate the survival rate by gender print(data.groupby('Sex')['Survived'].mean())
```

output:



#### 4.2.7

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked\_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

### Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|             |          |        |      |     |     |       |       |        |      |       |          |

**Note:** Refer to the visible test case for better reference.

```
Code: import pandas
as pd import numpy
as np
```

```
# Load the Titanic dataset data = pd.read_csv('Titanic-Dataset.csv') data['FamilySize']
= data['SibSp'] + data['Parch'] data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

# 1. Calculate the survival rate by class `print(data.groupby('Pclass')['Survived'].mean())`

# 2. Calculate the survival rate by embarked location

`print(data.groupby('Embarked_S')['Survived'].mean())`

# 3. Calculate the survival rate by family size

`print(data.groupby('FamilySize')['Survived'].mean())`

# 4. Calculate the survival rate by being alone

`print(data.groupby('IsAlone')['Survived'].mean())` # 5. Get the average fare by class

`print(data.groupby('Pclass')['Fare'].mean())`

# 6. Get the average age by class `print(data.groupby('Pclass')['Age'].mean())`

# 7. Get the average age by survival status `print(data.groupby('Survived')['Age'].mean())`

# 8. Get the average fare by survival status `print(data.groupby('Survived')['Fare'].mean())`

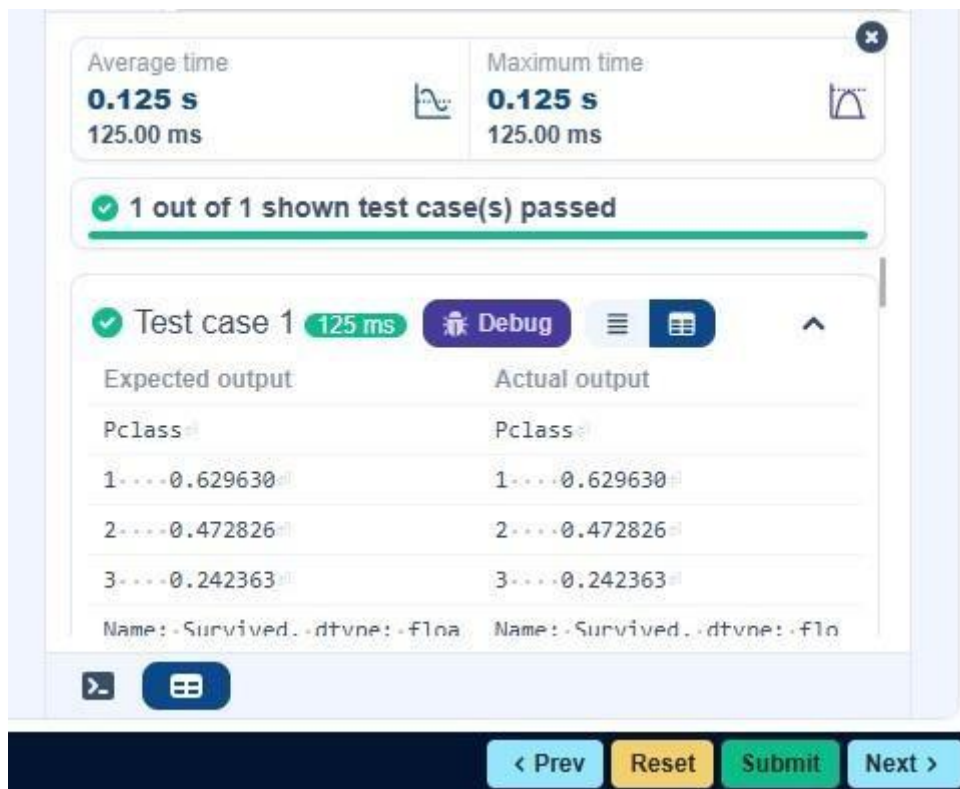
# 9. Get the number of survivors by class `print(data[data['Survived']`

`== 1]['Pclass'].value_counts())`

# 10. Get the number of non-survivors by class

`print(data[data['Survived'] == 0]['Pclass'].value_counts())`

output :



#### 4.2.8

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked\_S).
4. Get the number of non-survivors by embarkation location (Embarked\_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|
|-------------|----------|--------|------|-----|-----|-------|-------|--------|------|-------|----------|

### Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC  
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

**Note:** Refer to the visible test case for better reference.

Code: import pandas

as pd import numpy

as np

# Load the Titanic dataset data = pd.read\_csv('Titanic-Dataset.csv') data

= pd.get\_dummies(data, columns=['Embarked'], drop\_first=True)

# 1. Get the number of survivors by gender

```
print(data[data['Survived'] == 1]['Sex'].value_counts())
```

```
# 2. Get the number of non-survivors by gender print(data[data['Survived'] ==  
0]['Sex'].value_counts())
```

```
# 3. Get the number of survivors by embarked location print(data[data['Survived'] ==  
1]['Embarked_S'].value_counts())
```

```
# 4. Get the number of non-survivors by embarked location print(data[data['Survived'] ==  
0]['Embarked_S'].value_counts())
```

```
# 5. Calculate the percentage of children (Age < 18) who survived children  
= data[data['Age'] < 18] print(children['Survived'].mean())
```

```
# 6. Calculate the percentage of adults (Age >= 18) who survived adults  
= data[data['Age'] >= 18] print(adults['Survived'].mean())
```

```
# 7. Get the median age of survivors print(data[data['Survived']  
== 1]['Age'].dropna().median())
```

```
# 8. Get the median age of non-survivors print(data[data['Survived']  
== 0]['Age'].dropna().median())
```

```
# 9. Get the median fare of survivors print(data[data['Survived'] ==  
1]['Fare'].dropna().median()) # 10. Get the median fare of non-  
survivors print(data[data['Survived'] ==  
0]['Fare'].dropna().median())
```



```
ticket,
7.25,
yer)"
310128
emale,
05,,S

625,E
09,21
g)",fe
,1,0,
```

36

Average time  
**0.082 s**  
82.00 ms

Maximum time  
**0.082 s**  
82.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ Test case 1 82 ms

Debug

^

| Expected output         | Actual output           |
|-------------------------|-------------------------|
| female....233           | female...233            |
| male.....109            | male.....109            |
| Name: Sex, dtype: int64 | Name: Sex, dtype: int64 |
| male.....468            | male.....468            |
| female.....81           | female.....81           |

>

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